



AUSTRIAN
ACADEMY OF
SCIENCES



Intro to CBS4DH

Massimiliano Carloni

2025-03-05

136142 KU Computational Background Skills for Digital Humanities (2025S)



Massimiliano Carloni

- Background in Classical Philology
- Since 2021 at ACDH-CH
- Data modeling and data preservation
- Currently:
 - Data repository ARCHE
 - Vocabulary Service
 - EU-funded project ATRIUM

Index

- How this course works
- Get to know each other
- CBS4DH: what does that mean?
- Three values in DH
 - Collaboration
 - Openness
 - Versatility

How this course works

- Six parts
 - Command line (5.-6.3.)
 - File formats and encodings (6.3.)
 - Git (19.3.)
 - Regular expressions (20.3.)
 - Minimal introduction to web technologies (26.3.)
 - Tools and resources for Digital Humanities (27.3.)
- Course material: https://github.com/acdh-oeaw/Teaching_CBS4DH/tree/2025S



How this course works

- **Attendance** is extremely important!
- Please **follow along** with the tutorials
- **Ask questions**
- **Check** if you have installed everything
 - Visual Studio Code: <https://code.visualstudio.com/>
 - (only Windows users) Gitbash: <https://gitforwindows.org/>
 - Create a GitHub account: <https://github.com/join>

Two questions for you

- What are you studying / have you studied previously?
- What operating system do you use?

CBS4DH

Computational Background Skills
for
Digital Humanities

CBS4DH

Computational Background Skills
for
Digital Humanities

What is Digital Humanities?

- The application of digital methods to research in the Humanities
- Varying degree of use of digital methods
- <https://whatisdigitalhumanities.com>

What Is Digital Humanities?

New quote

- Around 800 definitions of DH
- [Raw data](#)

What Is Digital Humanities?

New quote

The best way I've found to describe what Digital Humanities is, in first place to myself, is using a quote from Thomas Pynchon's Crying of Lot 49: 'You know what a miracle is ... another world's intrusion into this one' **Federico Meschini, De Montfort University**

NB: Click the New quote button to display a new random quote. Quotes were pulled from participants from the Day of DH between 2009-2014. As of January 2015, the database contains 817 rows and randomly selects a quote each time the page is loaded. If you want to do something cool with the data, I am providing the code and data here.

Made by Jason Heppler. Problems? Questions? I'm @jaheppler on Twitter.

Not only Humanities

- Social Sciences and Humanities (**SSH**)
 - For example, SSHOC ([Social Sciences and Humanities Open Cloud/Cluster](#))
- Political science
- Business
- Broad application of certain techniques/research areas
 - e.g. Natural Language Processing
- And many others

CBS4DH

Computational Background Skills
for
Digital Humanities

CBS4DH

Computational Background Skills
for
Digital Humanities

„Push your changes to the repo“

CSS

```
^[a-zA-Z0-9._%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$
```

HTML

„You can write a shell script to do this“

131.130.70.8

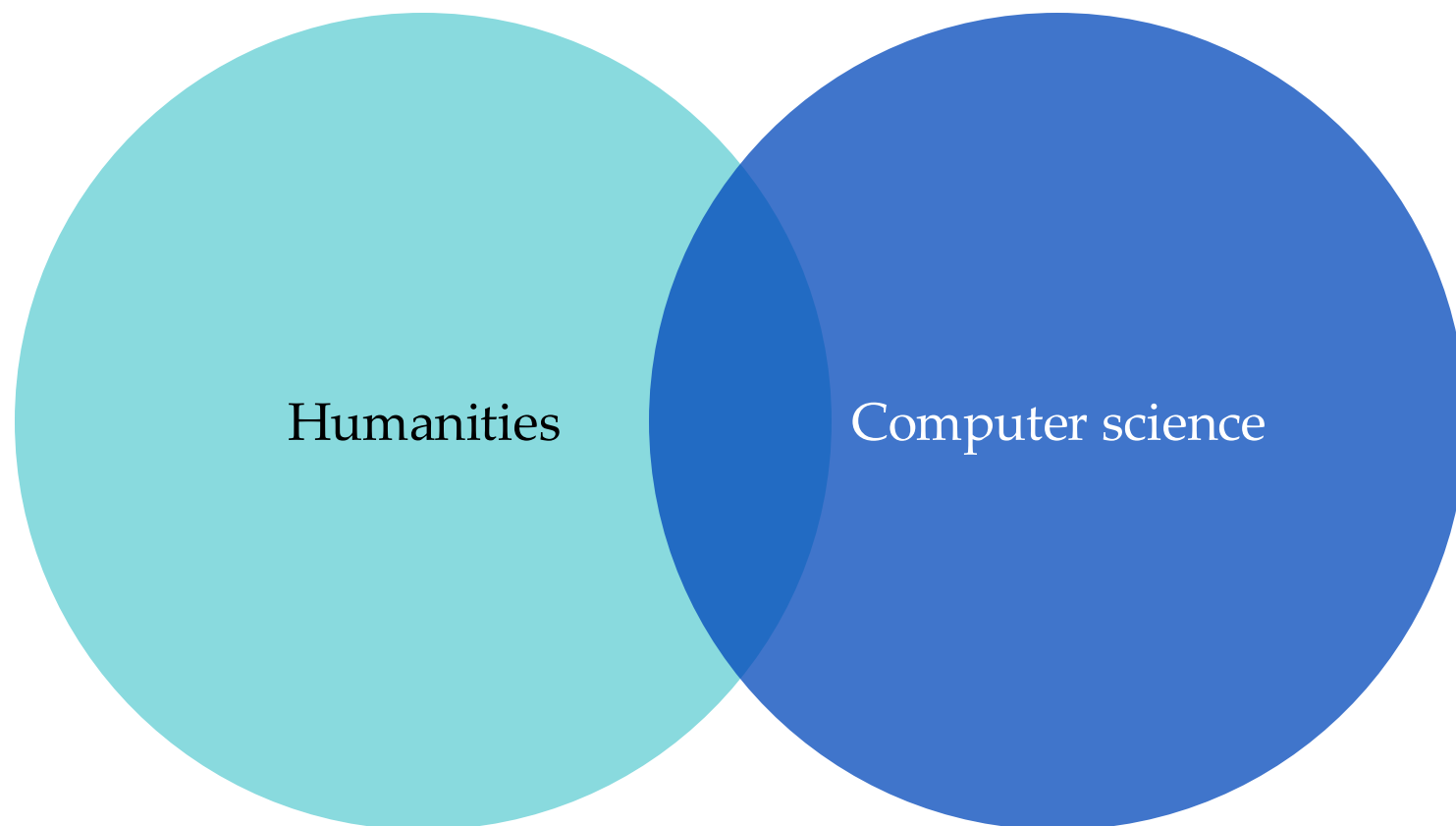
```
Alice,25,New York,Engineer
Bob,30,Los Angeles,Doctor
Charlie,35,Chicago,Artist
```

CBS4DH

Computational Background Skills
for
Digital Humanities

CBS4DH

Computational Background Skills
for
Digital Humanities



Some things you can do with these skills

- Print a list of all files in a directory
- Find all instances of “colour” and “color” with a single search
- Keep a change history of your document
- Create a basic website
- Find new opportunities to learn
- ... and many more!

Three ‚values‘ in DH

Collaboration

Communities and collaboration

- DH project = no one-man show!
- Collaboration between different actors, stakeholders and communities
 - Digital Humanists
 - Developers
 - More “traditional” scholars
 - Different departments / institutions
 - Broader audience



One Man Band, CC BY 2.0, Created by [Jarekt](#)

Digital Humanities Centers / Institutes

- [Digital Humanities @ University of Vienna](#)
- [Austrian Centre for Digital Humanities and Cultural Heritage](#) (at ÖAW)
- University of Graz – [Institut für digitale Geisteswissenschaften](#)
- Lists
 - [EADH Digital Humanities Centers](#)
 - [NYU List of Digital Humanities Centers](#)

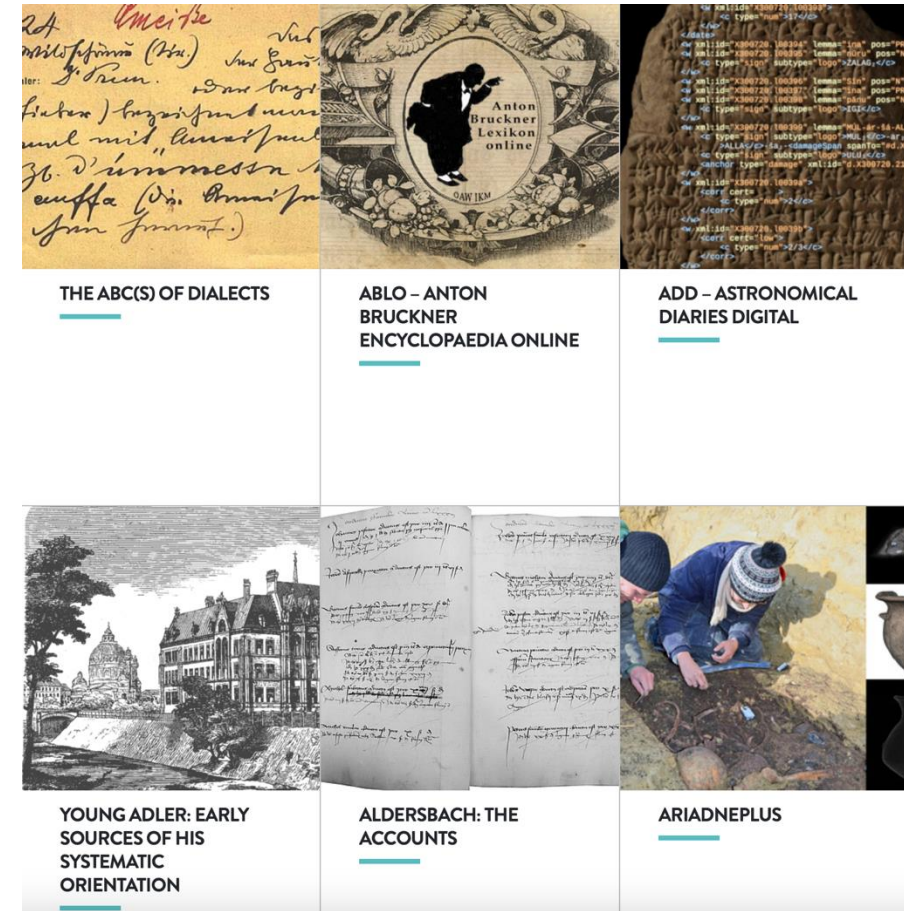
Austrian Centre for Digital Humanities and Cultural Heritage (ACDH-CH)

- Founded in 2015, concept from 2013
- Mission: *fostering the use of digital methods and tools in the humanities research in Austria*
- Rapid growth
 - From 12 in 2015 to 141 in 2024
 - ~95 [projects](#)
- Since 2020 + “Cultural Heritage”



Austrian Centre for Digital Humanities and Cultural Heritage (ACDH-CH)

- Four research units
 - Linguistics
 - Literature and Text
 - Musicology
 - DHRI – Digital Humanities Research & Infrastructure
- Both a research *and* a service-providing institution (What we offer)



DH associations

- DHd – Digital Humanities im deutschsprachigen Raum
<https://digitalhumanities.de>
- EADH – European Association for Digital Humanities
<http://eadh.org>
- ADHO – Alliance of Digital Humanities Organizations
<https://adho.org> (list of partners: <https://adho.org/membership-individual/>)
- MLA2S – Machine Learning at the Austrian Academy of Sciences
<http://www.oeaw.ac.at/mla2s>

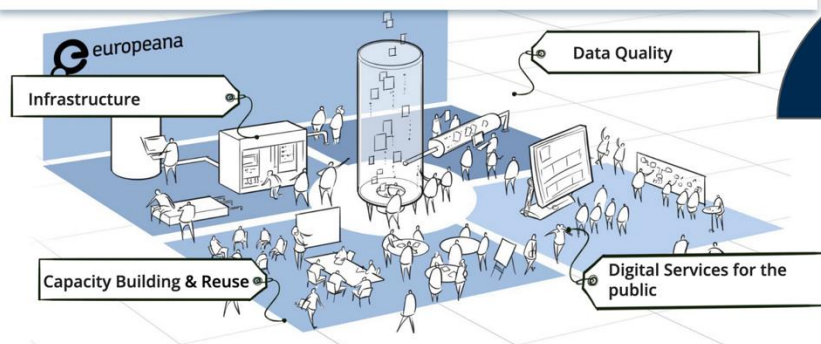


Research Infrastructures

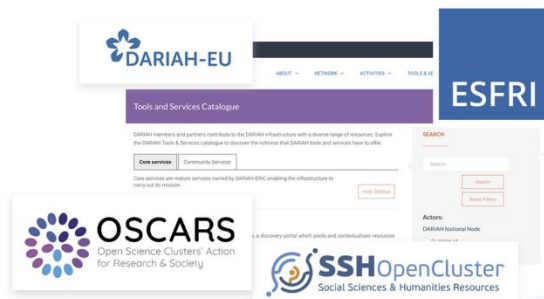
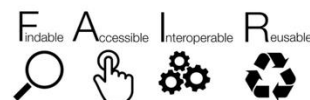
- “facilities that provide resources and services for research communities to conduct research and foster innovation” ([European Commission](#))
 - No humans mentioned!
- Networks of people and expertise
- Both bottom-up and top-down
 - Exchange of ideas and knowledge sharing
 - Consolidation, harmonization (standards, best practices)
 - Provision of services

The European Research Landscape

**Role: Interoperable Data Aggregation
(beyond metadata)**



**Role: State of the Art processing of
Cultural Heritage Data (e.g AI-based)**



Social Sciences & Humanities Open Marketplace

Discover new and contextualised resources for your research in Social Sciences and Humanities: tools, services, training materials, workflows and datasets. Read more...

All categories Search



**Role: Publishing of tools; methods
and derived datasets for reuse**



DARIAH

- European *Digital Research Infrastructure for the Arts and Humanities*
- Strong voice for the DH community on the European level
- Lines of action
 - Common policies & guidelines
 - Dissemination and discoverability of resources ([SSH Open Marketplace](#))
 - Knowledge sharing ([DARIAH Campus](#))
 - Integration with other initiatives, e.g. European Open Science Cloud ([EOSC](#))



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Discover new and contextualised resources for your research in Social Sciences and Humanities: tools, services, training materials, workflows and datasets. [Read more...](#)

All categories



Search

[SSH Open
Marketplace](#)

[Search](#)[Home](#) / [Tools & services](#) / [Gephi](#)

Gephi

[Copy to clipboard](#)

Gephi is the leading visualization and exploration software for all kinds of graphs and networks.

Gephi is open-source and free. Runs on Windows, Mac OS X and Linux.

[Go to Tool or service](#)

Details

ACCESS

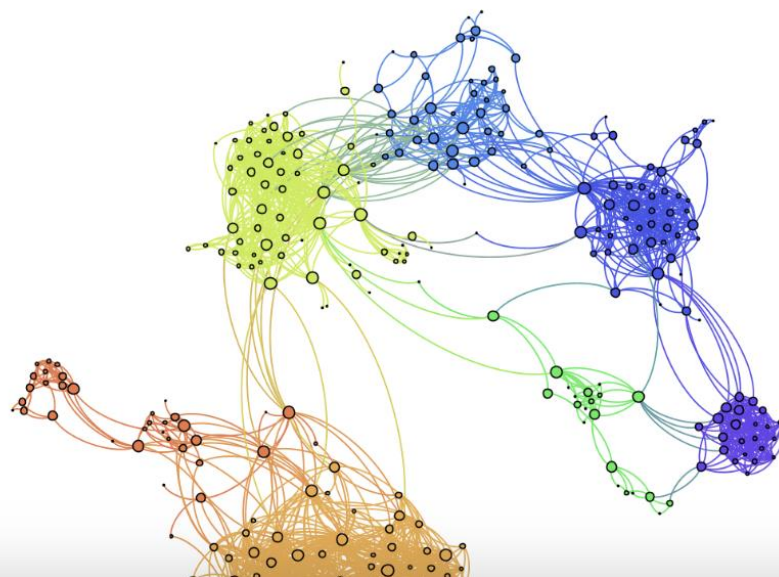
License [Common Development and Distribution License 1.0](#) [GNU General Public License v3.0 only](#)

Terms Of Use [Free](#)

CATEGORISATION

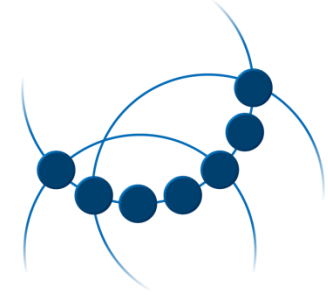
Activity [Capturing](#) [Modeling](#) [Network Analysis](#) [Analyzing](#) [Visual Analysis](#) [Data Visualization](#) [Relational Analysis](#) [Spatial Analysis](#) [Content Analysis](#) [Discovering](#)

Keyword [Models](#) [graphs](#) [french](#) [R](#) [faceted browser](#) [graph streaming](#) [recommended](#)



CLARIN

CLARIN
Common Language Resources and
Technology Infrastructure



- *Common Language Resources and Technology Infrastructure*
- Easy and sustainable access to a broad range of language data and tools
- Distributed research infrastructure, with [centers all over Europe](#)
- Services
 - [Content search](#) (search for specific patterns across collections)
 - [Virtual Language Observatory](#) (explore language resources and tools)
 - [Language Resource Switchboard](#) (process your data with specialized tools)
 - [Learning Hub](#) and [DH Course Registry](#) (find learning opportunities in DH)

DIGITAL HUMANITIES COURSE REGISTRY



INFO

☰

🏠

FILTER

SORT

Find a course directly by name or university

NAME	UNIVERSITY	PLACE	DATE	TYPE
MA in Digital Humanities	King's College London	London, United Kingdom	24 Sept ↻ 1 years	Master Programme
MSc in Digital Scholarship	University of Oxford	Oxford, United Kingdom	9 Oct ↻ 11 months	Master Programme
Digital Stylometry	Freie Universität Berlin	Berlin, Germany	17 Apr 2025 1 semester	Course
Introduction to the Digital Humanities	Freie Universität Berlin	Berlin, Germany	14 Apr ↻ 1 semester	Course
Postgraduate Certificate in Digital Humanities and Culture	Trinity College Dublin	Dublin, Ireland	5 Sept 2025 2 semester	Course
Postgraduate Diploma in Digital Humanities & Culture	Trinity College Dublin	Dublin, Ireland	5 Sept ↻ 2 semester	Course
MPhil in Digital Humanities and Culture	Trinity College Dublin	Dublin, Ireland	5 Sept ↻ 1 years	Master Programme
Swedish National Research School in Digital Humanities: Dat ...	Uppsala University	Uppsala, Sweden	25 Sept ↻ 12 months	PhD Programme
Digital Humanities	Higher School of Economics Perm	Perm, Russian Federation	1 Sept ↻ 2 years	Master Programme
Language and computers	Universitetet i Bergen	Bergen, Norway	15 Jan ↻ 1 semester	Bachelor Programme
Master in Textual Heritage & Digital Humanities [Máster en ...	Universidad de Salamanca	Salamanca, Spain	23 Sept ↻ 2 semester	Master Programme
Corpus-Driven Lexicology	Mykolas Romeris University	Vilnius, Lithuania	1 Feb ↻ 5 months	Bachelor Programme



DH Course Registry



Austrian DH landscape

- CLARIAH-AT – clariah.at
 - CLARIN + DARIAH at the Austrian level
- DiTAH – ditah.at
 - *Digitale Transformation der österreichischen Geisteswissenschaften*
 - Four-year project (2020-2024) for the establishment of Digital Humanities in Austria
- Funding schemes (e.g. CLARIAH-AT [Project funding](#), [Funding for Junior Researchers](#), [Transkribus Scholarship](#))
- Next [CLARIN Annual Conference 2025](#) will take place in Vienna!
(30 September – 2 October 2025)



CLARIAH-AT Partners



University for
Continuing
Education Krems



UNIVERSITÄT GRAZ
UNIVERSITY OF GRAZ



Cultural Heritage (CH) Infrastructures

- Improve the availability of CH objects for research and beyond
- Particularly relevant for [GLAM](#) sector (Galleries, Libraries, Archives and Museums)
- Austrian level: [Kulturpool](#)
- European level: [Europeana](#)
- ECCCH
([European Collaborative Cloud for Cultural Heritage](#))





... and many others

[Full list of Europeana aggregators](#)



Slovakiana



Swedish Open Cultural
Heritage



CulturaItalia



National Heritage Institute,
Bucharest



Greek Aggregator
SearchCulture.gr



Gallica



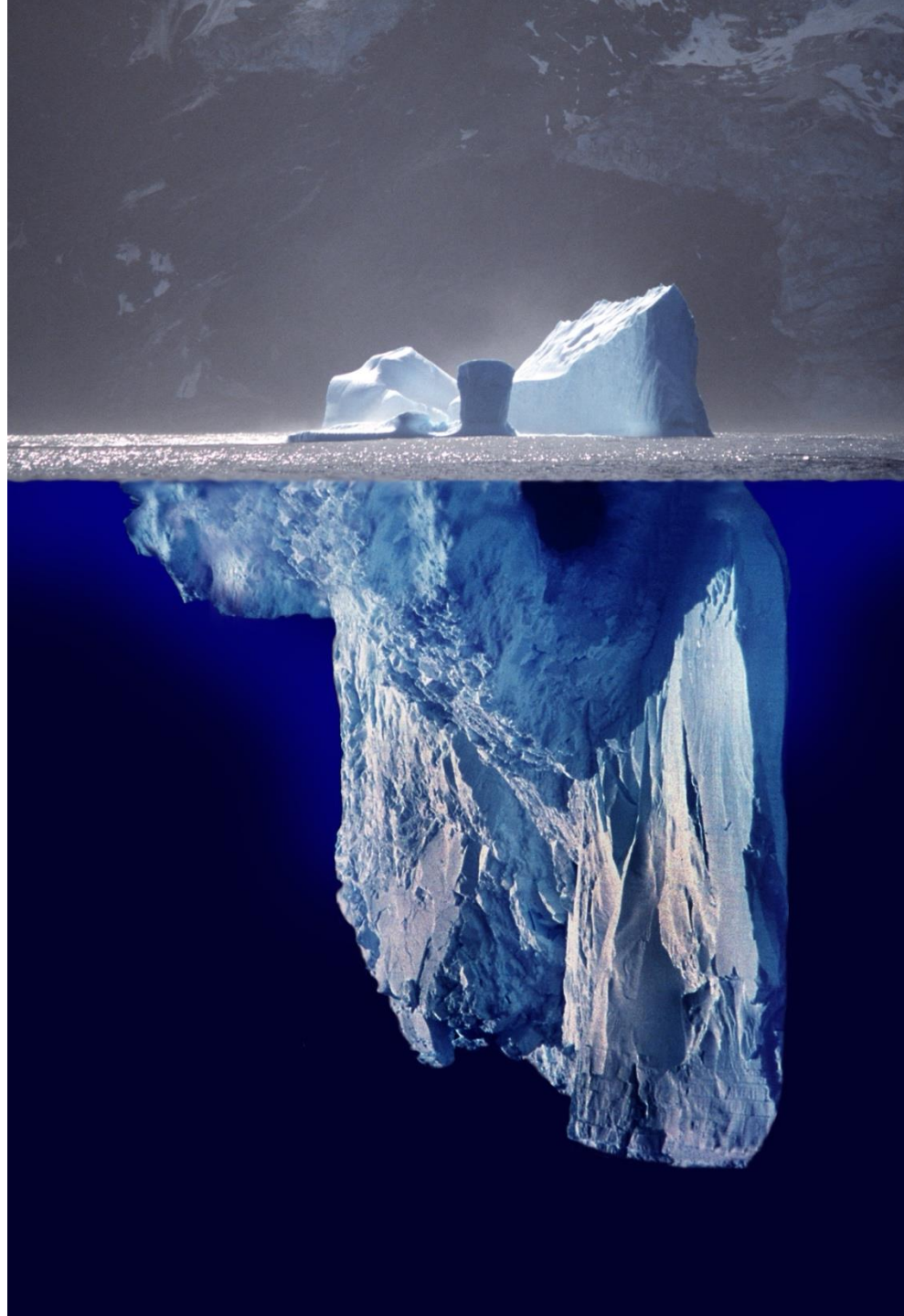
Openness

Website
Book/Article
Digital edition

Below the surface...

Data

Scans
Photos
Database
3D models
Transcriptions
Audio recordings
Video recordings
Reports and notes
...



CC BY-SA 3.0, created by Uwe Kils

Data is/are important

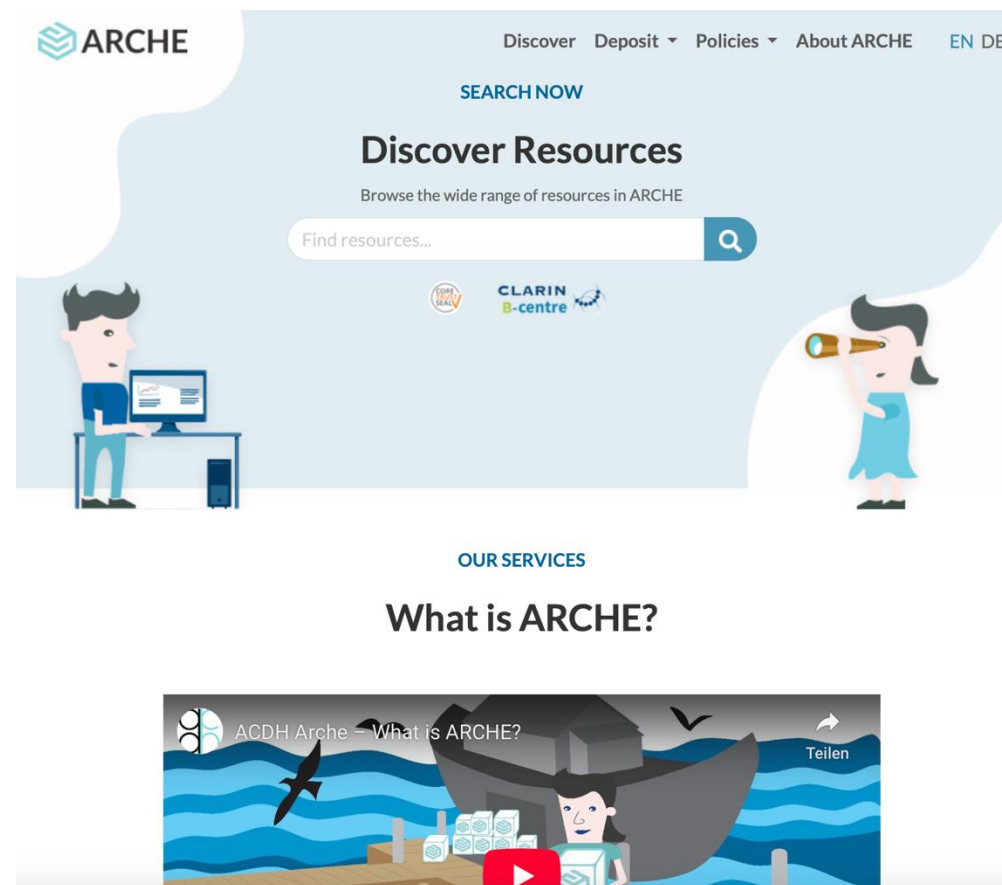
- Data is the backbone of your research, even in the Humanities
- Open data allows you to validate your results
- Open data allows others to build on your research
- **Data first!** Research data as one of the most important outputs of projects
 - Datasets as first-class citizens in the research landscape
 - Can be published in repositories like [Zenodo](#) ...
 - ... and referenced as publications (e.g. with a [DOI](#))

zenodo



ARCHE

- Repository for digital research data in the Humanities
- Developed and hosted by ACDH-CH
- Curation team that helps in deposition process, checks & dissemination
- Integration with CLARIN VLO and other European catalogues



Open Data

- FAIR Principles
 - Findability
 - Accessibility
 - Interoperability
 - Reusability
- Governments and local authorities
 - <https://www.data.gov.uk>
 - <https://www.data.gv.at>

data.gv.at

56072 Datensätze
770 Anwendungen
1356 Organisationen

de en

Umfrage zur Verbesserung von data.gv.at gestartet

Daten > Anwendungen > Infos > News >

Suche Suchbegriff ... erweiterte Suche

Themen

- Bevölkerung und Gesellschaft
- Bildung, Kultur und Sport
- Energie
- Gesundheit
- Internationale Themen
- Justiz, Rechtssystem und öffentliche Sicherheit
- Landwirtschaft, Fischerei, Forstwirtschaft und Nahrungsmittel
- Regierung und öffentlicher Sektor
- Regionen und Städte
- Verkehr
- Umwelt
- Wirtschaft und Finanzen



“Openness” in modern trends


[Join us](#)
[Sign in](#)

EMERGING TECHNOLOGIES

What is open-source AI and how could DeepSeek change the industry?

Feb 5, 2025



Bundesministerium
Bildung, Wissenschaft
und Forschung



Home > ... > Strategien > Open Innovation Strategie

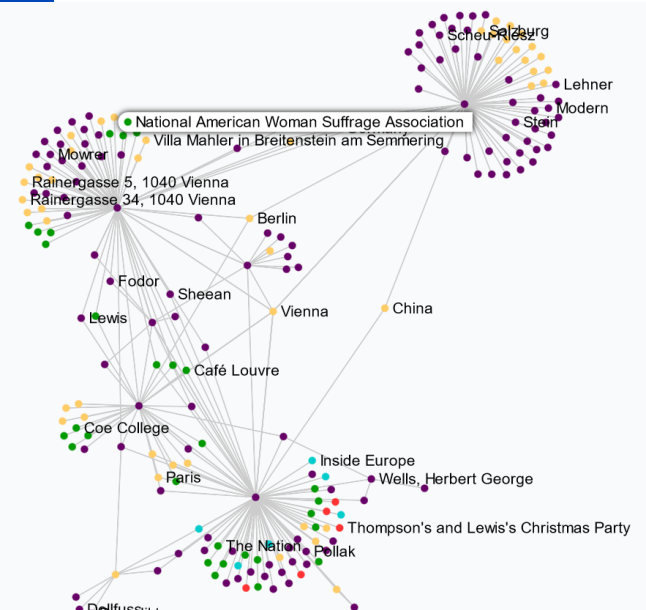
Open Innovation Strategie

Österreich ist der erste europäische Mitgliedsstaat, der eine umfassende nationale Open Innovation Strategie entwickelt hat.

Im Zuge eines breiten Beteiligungsprozesses, in welchem nicht nur Stakeholder aus Forschung, Wissenschaft und Wirtschaft sondern auch die Zivilbevölkerung eingebunden waren, wurde im Jahr 2016 eine Open Innovation Strategie für Österreich erstellt. Diese umfasst neben einer Analyse des nationalen und internationalen Status Quo zum Thema Open Innovation auch eine Vision für 2025 und konkrete Maßnahmen zur Umsetzung.



Versatility



in sehr guter Maske.
14/11 Vm. bei Gisa. - Telegr. an Hauptmann,
- Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten
nicht gesehen; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler un
anderer Sohn. Erzählt mir von seinen Krankheiten und Operat
Über Antisemitismus u. dergl., amerik. Verhältnisse.-
Forest in der Tram; der sehr entzückt von Heini sprach.-
In der N. W. Bühne, wegen Sitzen zu W. L.;- spreche fl

Workflows

Create a dictionary in TEI

Workflow steps (5)

- 1 Build the model of the dictionary
- 2 Create a corpus of useful resources for the dictionary.
- 3 Setting up the editing environment
- 4 Setting up the publishing environment
- 5 Provide for long-term availability

Related items (4)

Dictionary Writing Systems \& Corpus Query Systems

No description provided.

[Read more](#)

BaseX Documentation

BaseX is a light-weight, high-performance and scalable XML Database and an XQuery 3.1 Processor with full support for the W3C Update...

[Read more](#)

Introducing Lexonomy: an open-source dictionary writing and publishing system

This demo introduces Lexonomy: a free, open-source, web-based dictionary writing and publishing system. In...

[Read more](#)

OBL

SUCHE ÜBER DAS ÖBL KONTAKT

Streit, Andreas

Streit Andreas, Architekt. Geb. Habendorf, Böhmen (Stráž nad Nisou, Tschechien), 15. 7. 1840; gest. Reichenau (Reichenau an der Rax, NO), 20. 1. 1916; röm-kath.

Liberec

seat of a first-order administrative division (PPLA)

wirkte in

LIFELINE

1840 Stráž nad Nisou

1858 Prag

1861 München

1863 Wien

1869 Österreich

1871 Wien

1882 Wien

1883 Wien

1886 Wien

1894 Wien

1900 Wien

1902 Wien

1905 Wien

1910 Wien

1916 Reichenau an der Rax

region - 11609024

London, Ontario

Canada

populated place - 4050060

Inner London, England

United Kingdom

region - 12042196

Outer London, England

United Kingdom

region - 12042197

Greater London, England

ARTHUR SCHNITZLER Tagebuch

Zur Edition Kalender Einträge Personen Werke Orte

Rmdth. Leon, Gebildete Menschen, Heini spielte eine Nebenrolle in sehr guter Maske.

14/11 Vm. bei Gisa.- Telegr. an Hauptmann zu seinem 60.-

- Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten
gesehen; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler und
anderer Sohn. Erzählt mir von seinen Krankheiten und Operationen.
Über Antisemitismus u. dergl., amerik. Verhältnisse.-

Forest in der Tram; der sehr entzückt von Heini sprach.-

In der N. W. Bühne, wegen Sitzen zu W. L.;- spreche flüchtig Korff.-



Dienstag, 14. November 1922

14/11 Vm. bei Gisa.— Telegr. an Hauptmann zu seinem 60.—

— Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten nicht gesehen; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler und sein anderer Sohn. Erzählt mir von seinen Krankheiten und Operationen. Über Antisemitismus u. dergl., amerik. Verhältnisse.—

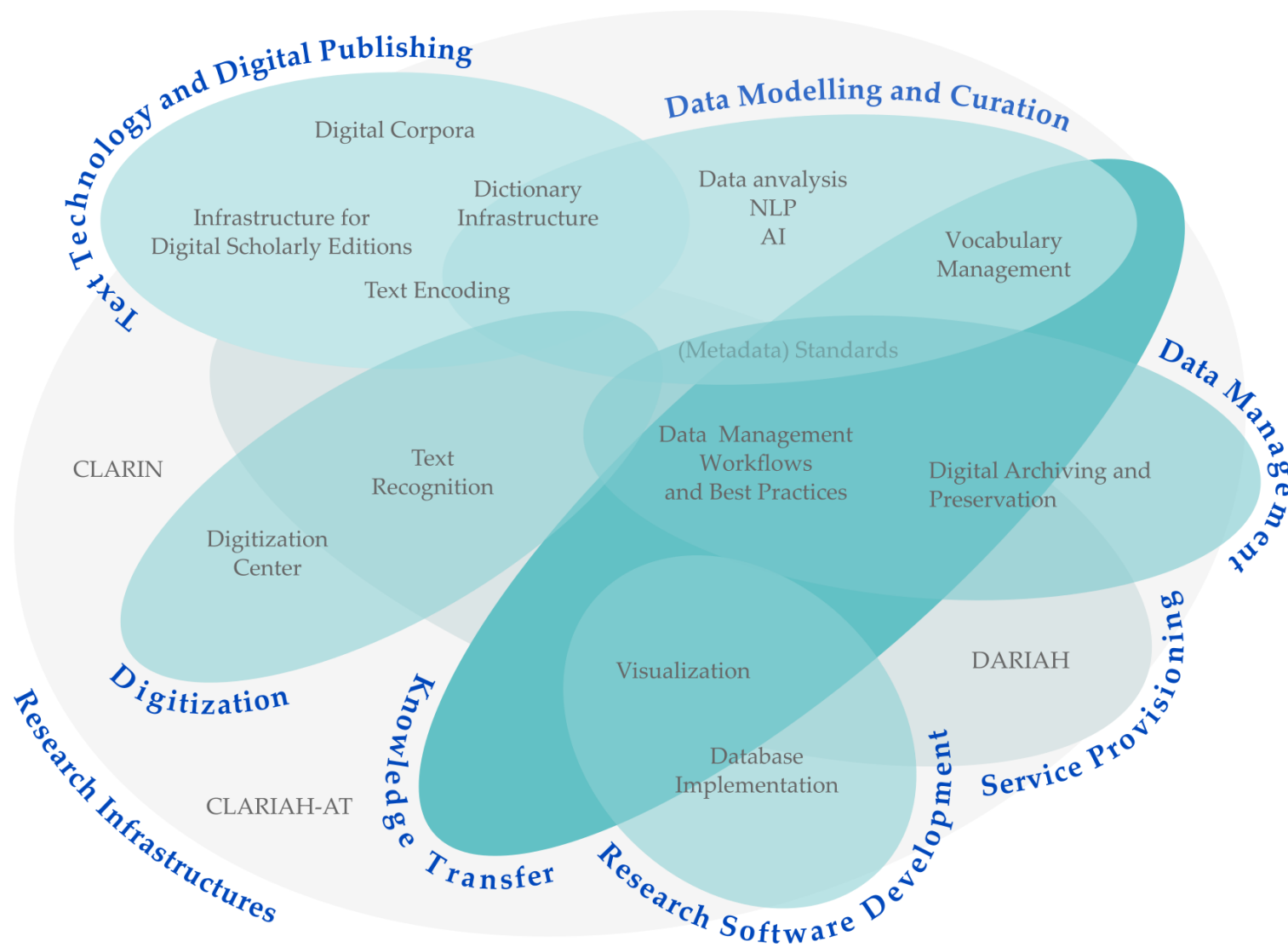
Forest in der Tram; der sehr entzückt von Heini sprach.—

In der N. W. Bühne, wegen Sitzen zu W. L.;- spreche flüchtig Korff.—

Mit Kolap Rmdth. „Weber“. Heini spielte den Kandidaten und den zweiten jungen Weber.

Gute Inszenierung von Martin. Das Stück wirkte wieder außerordentlich. Ein Gestalter

Areas of Expertise



Technology trends

Perceptron

Rosenblatt, F. *The Design of an Intelligent Automaton*, Research Reviews, Office of Naval Research. Washington, October 1958, 7.

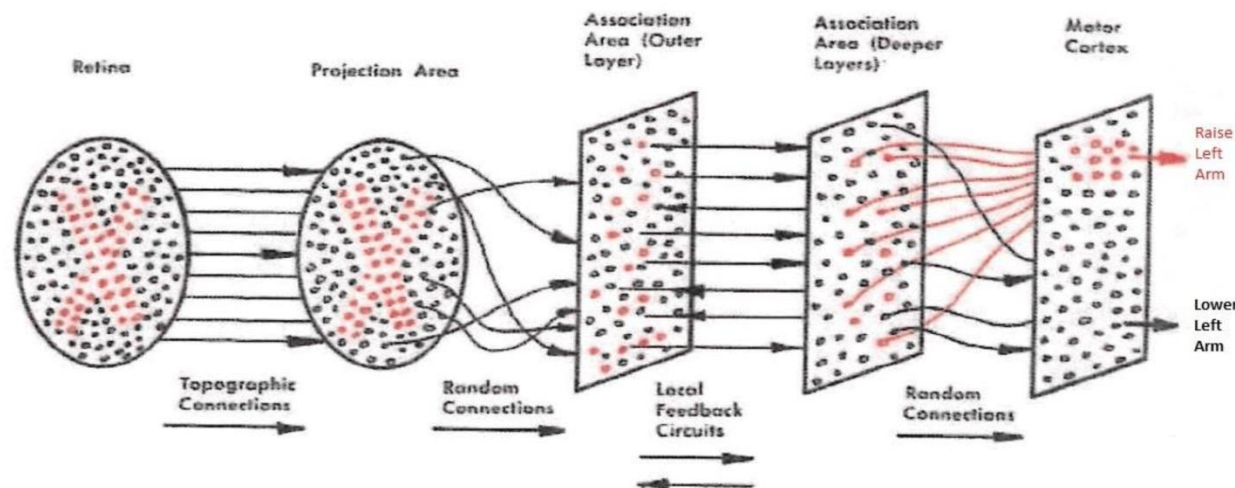


FIG. 1 — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

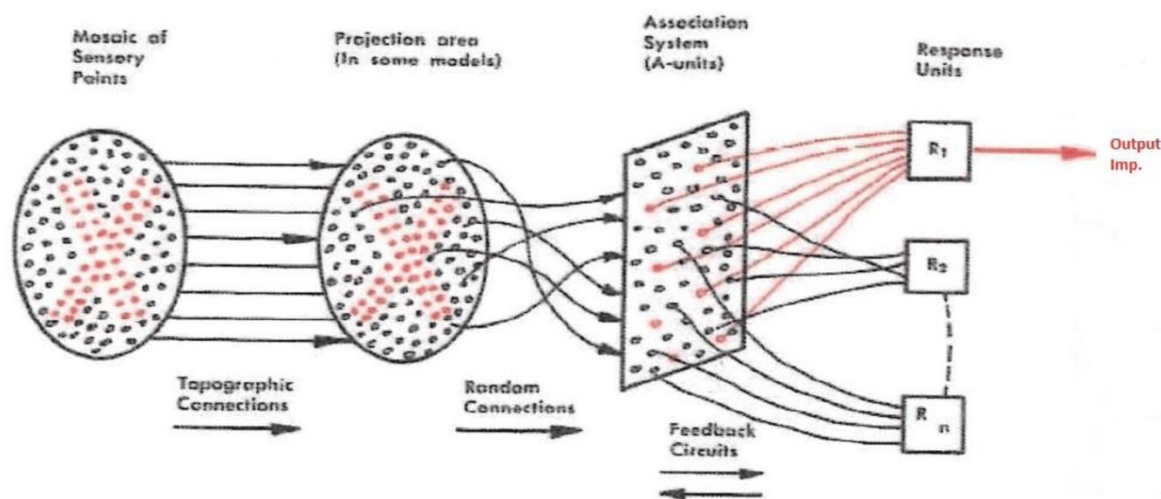
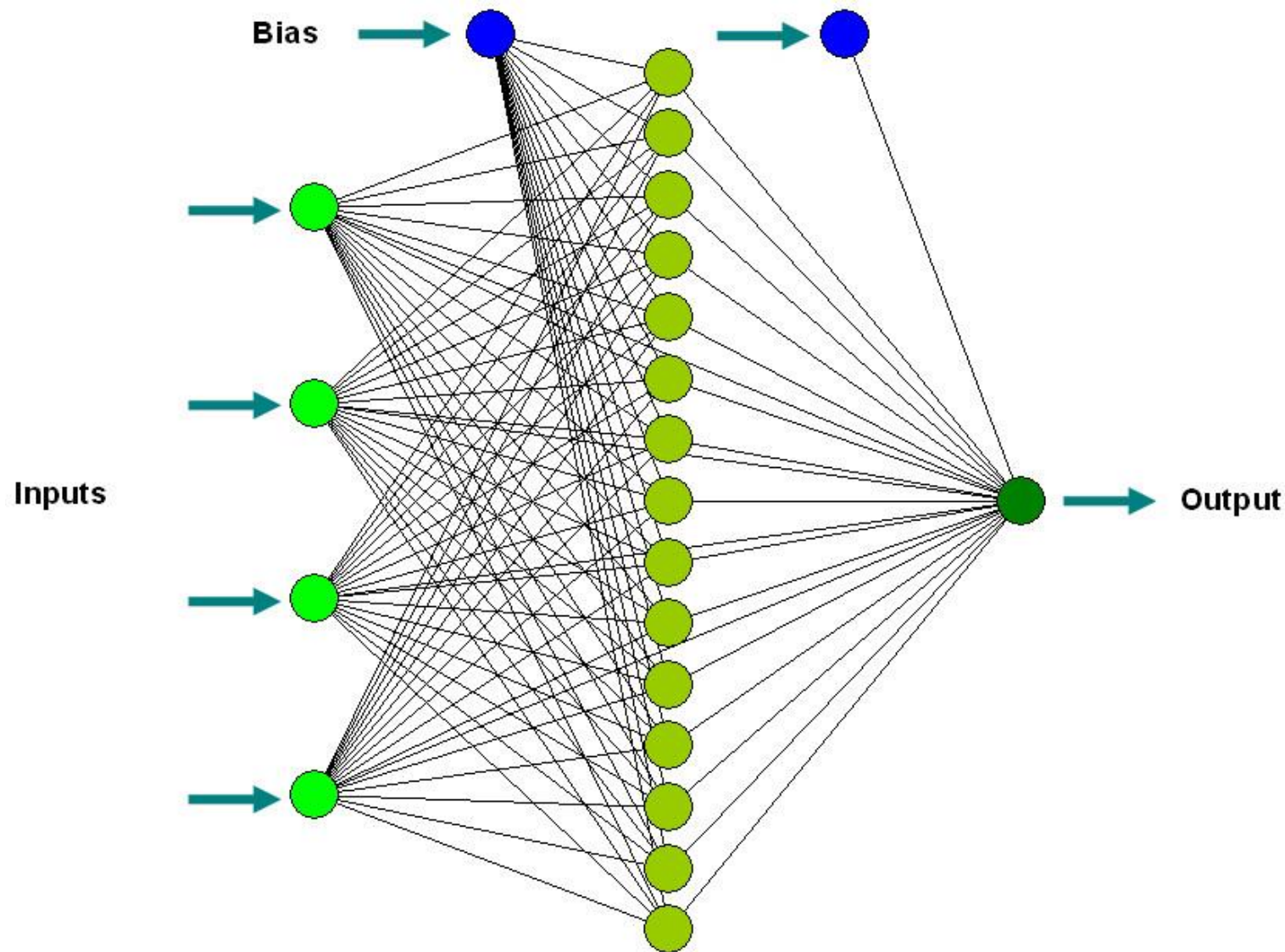


FIG. 2 — Organization of a perceptron.

Technology trends

Deep neural network

Basis for modern
Artificial
Intelligence



Technology trends



The Semantic Web

Scientific American,
May 2001
Volume 284, Issue 5

Technology trends

- Omnipresence of the Internet
- Mobile devices
- Virtual / Augmented Reality
- Widespread use of Artificial Intelligence (machine learning, esp. deep learning)
- Reflections on the consequences of technology



INDIGO Graffiti Project, [YouTube](#)

Artificial Intelligence

- Already being used in projects, mainly: HTR/OCR (Transkribus), NLP (spaCy)
- Many potential uses (many hitherto human tasks):
 - NLP (tagging, parsing; audio transcription; translation tasks)
 - Word Sense Disambiguation
 - AI for text encoding / annotating (editions)
 - Semantic Entity Linking
 - (Meta-)Data mining
 - Generative AI
- Towards combining stochastic and knowledge-based approaches

Training and keeping up-to-date

- Events
 - [ACDH-CH Events](#)
 - [ACDH-CH Event Series](#) (Tool Galleries, Research Days, ...)
 - University of Vienna DH Ringvorlesung ([2025S](#))
- Open Educational Resources (OER)
 - [DARIAH Campus](#)
 - [ACDH-CH How To](#)
- Internships ([at ACDH-CH](#))

Mailing lists

- Humanist: https://dhhumanist.org/membership_form.php
- DHd-Mailingliste: <https://lists.uni-hamburg.de/mailman/listinfo/dhd>
- Linguist: <https://linguistlist.org/subscribe/>
- Corpora: <https://list.elra.info/mailman3/postorius/lists/corpora.list.elra.info/>
- DH Summer Institute: <https://lists.uvic.ca/mailman/listinfo/institute>
- TLDR (generally on tech): <https://tldr.tech>
- The Batch (on AI): <https://www.deeplearning.ai/the-batch/>

Thank you! Questions?

<https://www.oeaw.ac.at/acdh>

<https://clariah.at>

<https://www.ditah.at>

<https://clarin.eu>

<https://darjah.eu>

<https://sshopencloud.eu>

Contact: massimiliano.carloni@oeaw.ac.at